



Physics-based Modelling of Radial Diffusion in the outer Van Allen belt



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INTRODUCTION

The outer Van Allen belt is a highly dynamic environment, with electrons reaching relativistic and ultra-relativistic energies (>10 MeV). Their acceleration is primarily attributed to (i) radial diffusion and (ii) local acceleration through wave-particle interactions. **Inward radial diffusion** transports electrons toward stronger magnetic fields, conserving the first two adiabatic invariants while violating the third.

In this work, we develop a **physics-based model** by numerically solving the **time-dependent Fokker-Planck equation** under constant adiabatic invariants μ and K , to investigate radial electron transport during geomagnetic storms. Here, we focus on the **September 2019** weak storm, which produced ~ 10 MeV electrons, mainly via radial diffusion.

DATA & METHODOLOGY

- **Van Allen Probes:** ECT electron differential fluxes & EMFISIS magnetic field measurements.
- **Himawari-8:** SEDA-e electron omnidirectional fluxes (*expansion of the model at GEO*)

For the numerical solution, we use **Radial Basis Function (RBF)** collocation methods, enabling comparison of different RBF formulations.

Governing equation

$$\frac{\partial f}{\partial t} = L^2 \left(\frac{D_{LL}}{L^2} \cdot \frac{\partial f}{\partial L} \right) - \frac{f}{\tau}$$

Discretization & Crank-Nicolson time integration

Meshless RBF collocation method $f(L, t^{n+1}) \approx \sum_{j=1}^N \lambda_j^{n+1} \varphi_j(r)$

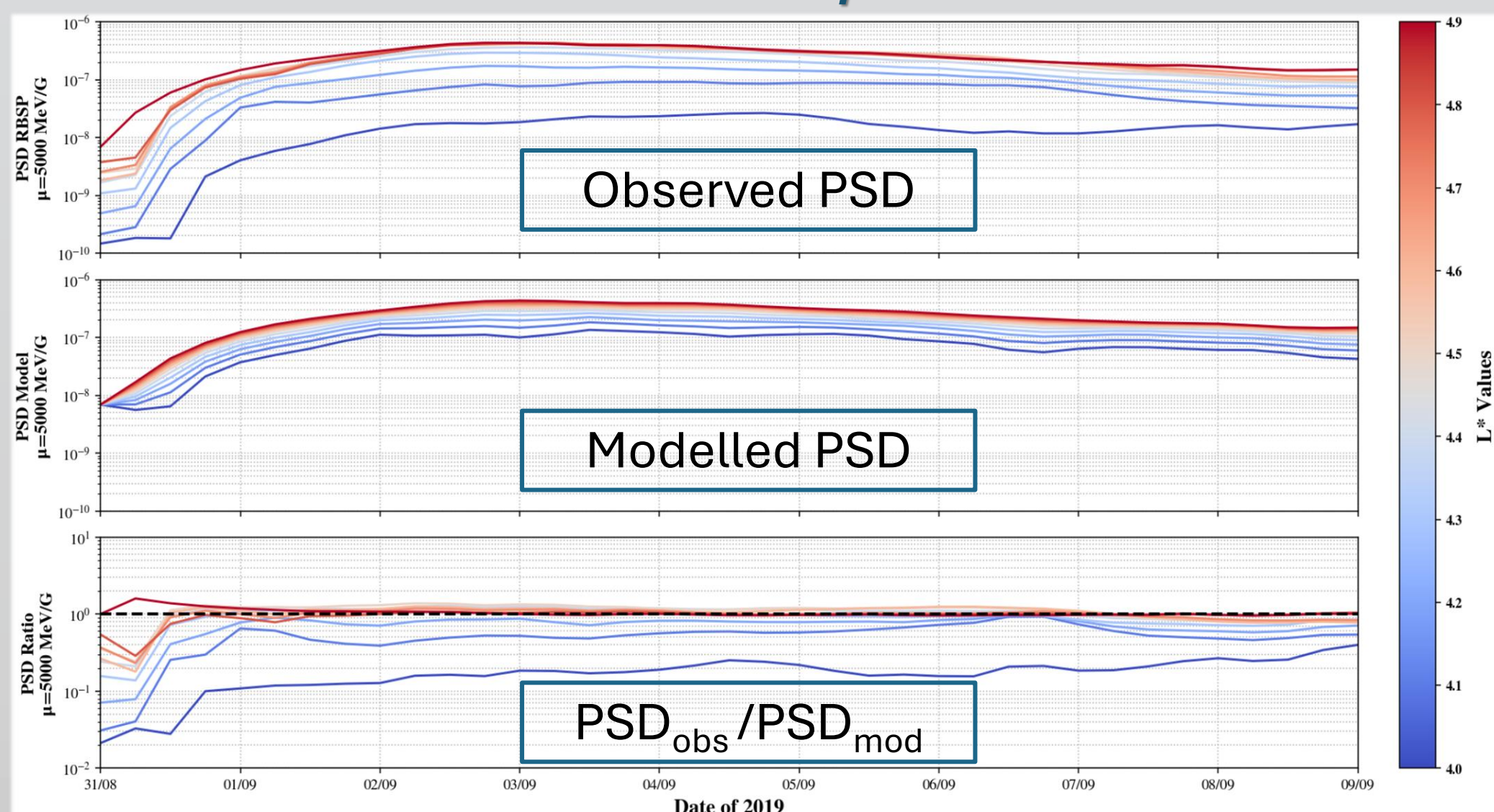
Model's Specifications

- $\tau \rightarrow \infty$ (negligible loss timescale)
- D_{LL} from the data-driven **EMERALD** model
- **Multiquadric RBF** $\varphi(r) = \sqrt{r^2 + c^2}$; $r = \|L - L_j\|$;
shape parameter $c = 6 \cdot 10^{-3}$
- **Grid:** $\Delta t = 6$ h; $\Delta L^* = 0.1$
- **Adiabatic invariants:** $\mu = 1000$ and 5000 MeV/G; $K = 0.03 G^{1/2} R_e$; $L^* = 4 - 5$, extended to 6.6 (Himawari-8 data). Values represent **relativistic & ultra-relativistic, near-equatorial mirroring electrons** in the **outer radiation belt**.

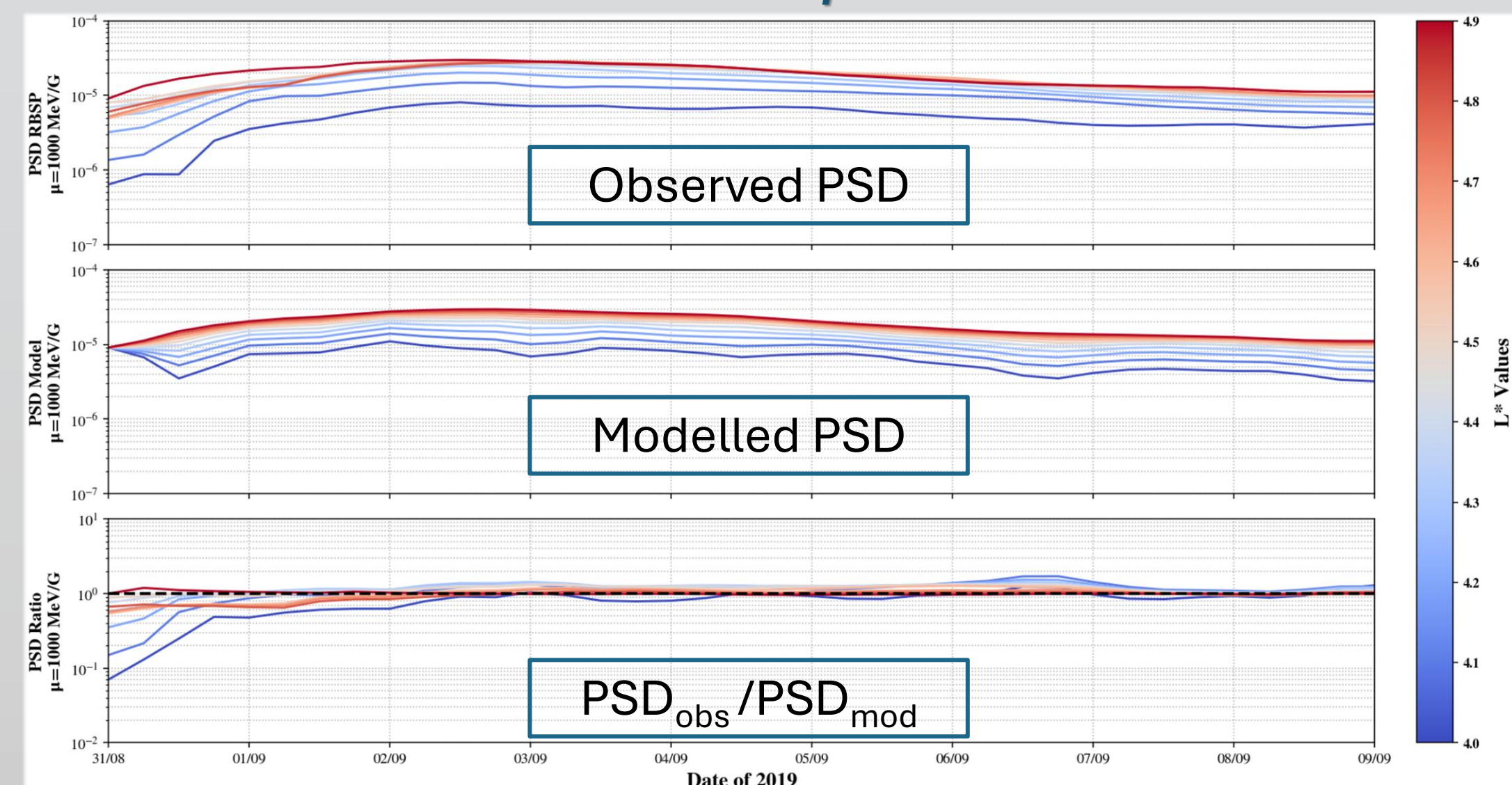
Data – dependency of the model

Observed PSD sets the **outer boundary** and the **initial condition** for all L^* values

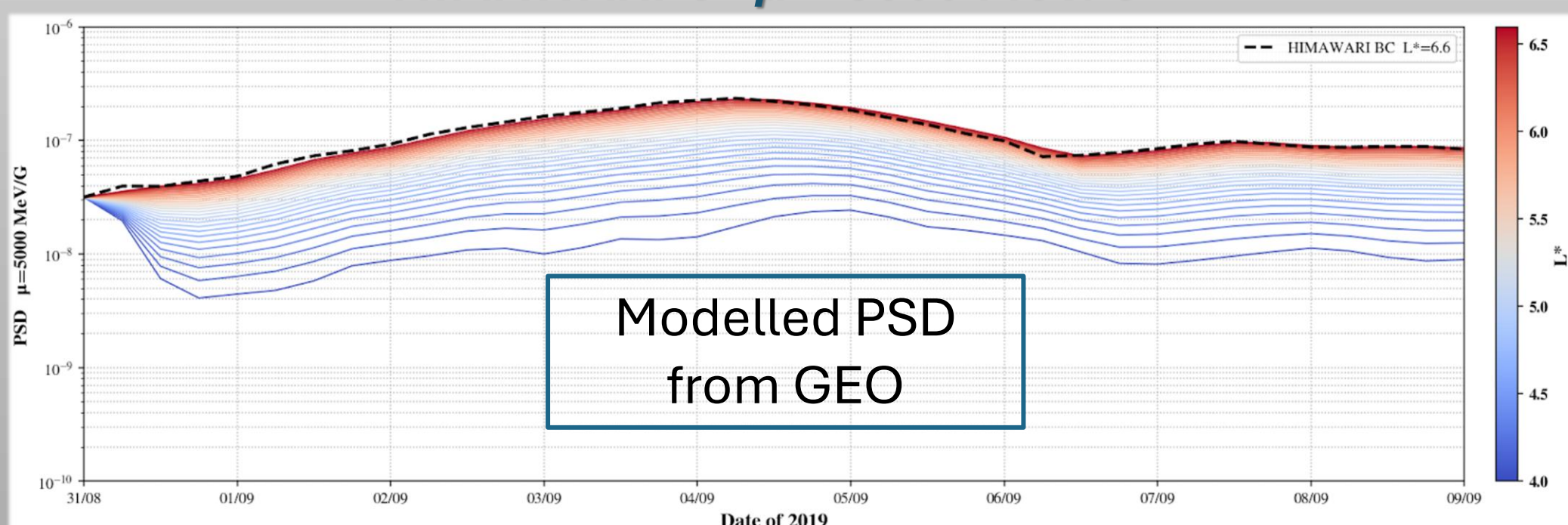
VAN ALLEN PROBES • $\mu = 5000$ MeV/G



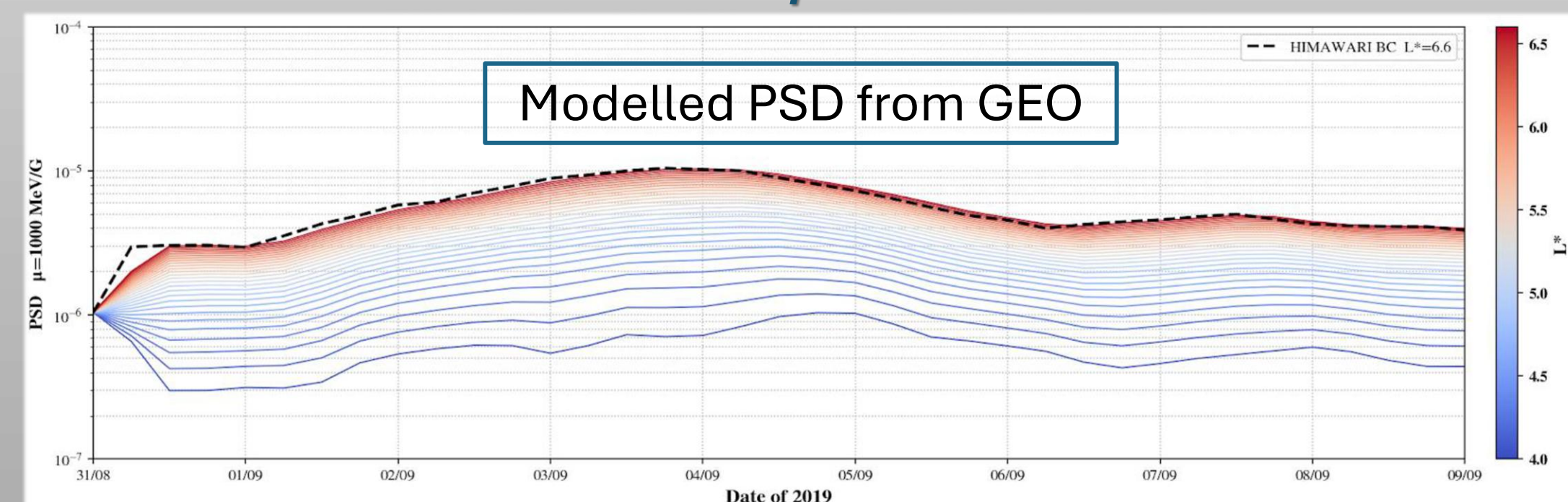
VAN ALLEN PROBES • $\mu = 1000$ MeV/G



HIMAWARI-8 • $\mu = 5000$ MeV/G



HIMAWARI-8 • $\mu = 1000$ MeV/G



CONCLUSIONS

- The model successfully reproduces the observed evolution of electron PSD, with deviations generally within a factor of ~ 3 during the storm's main phase.
- A **multiquadric RBF** is the most reliable; the optimal shape parameter can vary by event.
- For $\mu = 5000$ MeV/G, discrepancies at lower L^* are linked to the location and dynamics of the **plasmopause** ($L_{pp}^* \approx 4$), which modulates ULF wave propagation and radial diffusion efficiency.
- **Early-phase PSD overestimation** at low L^* arises primarily from simplified initial conditions and the finite response time of the diffusion model.
- The model captures the large-scale acceleration and transport of electrons, but cannot reproduce **localized enhancements** driven by wave-particle interactions.
- The **successful expansion** of the modelled domain to **GEO distances** ($L^* \approx 6.6$) highlights the potential of the approach as an **operational radiation belt nowcasting tool**, enabling continuous monitoring of electron dynamics across a larger portion of the outer belt while maintaining predictive skill.

More questions?

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